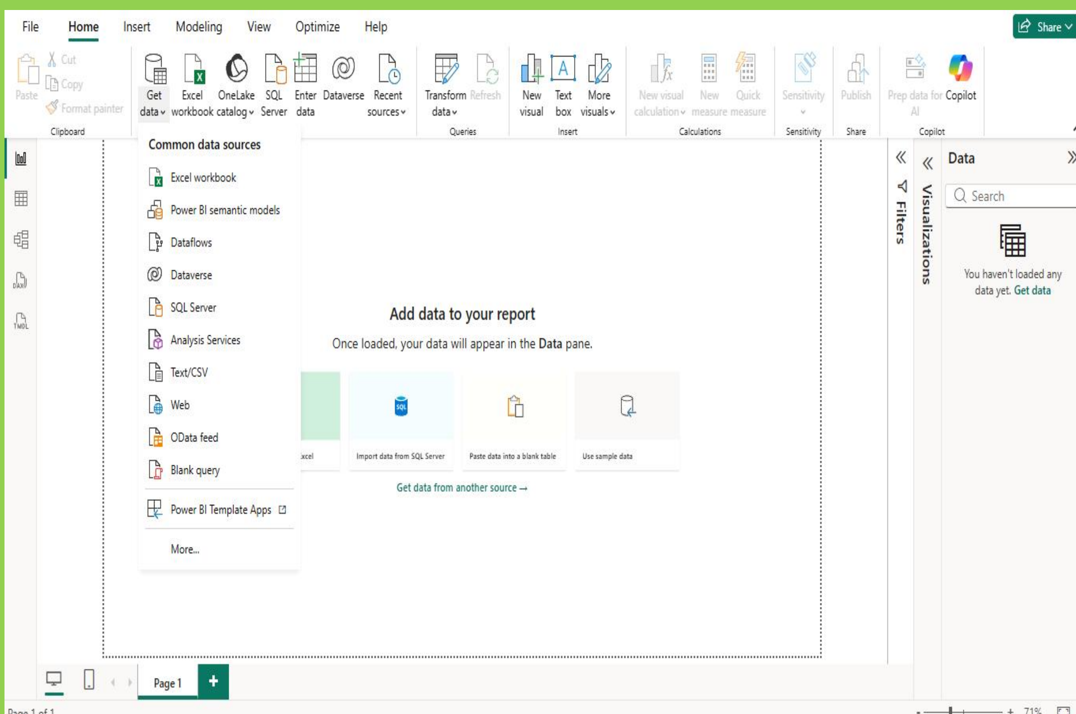
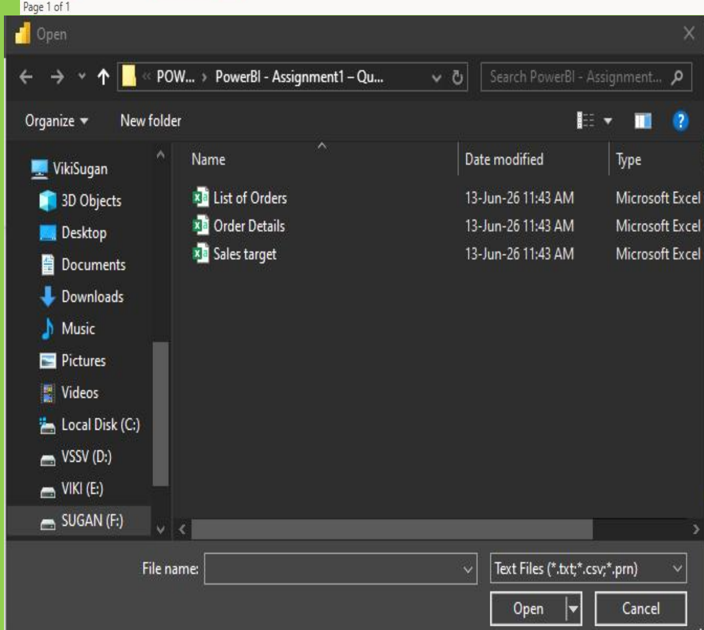
Tasks

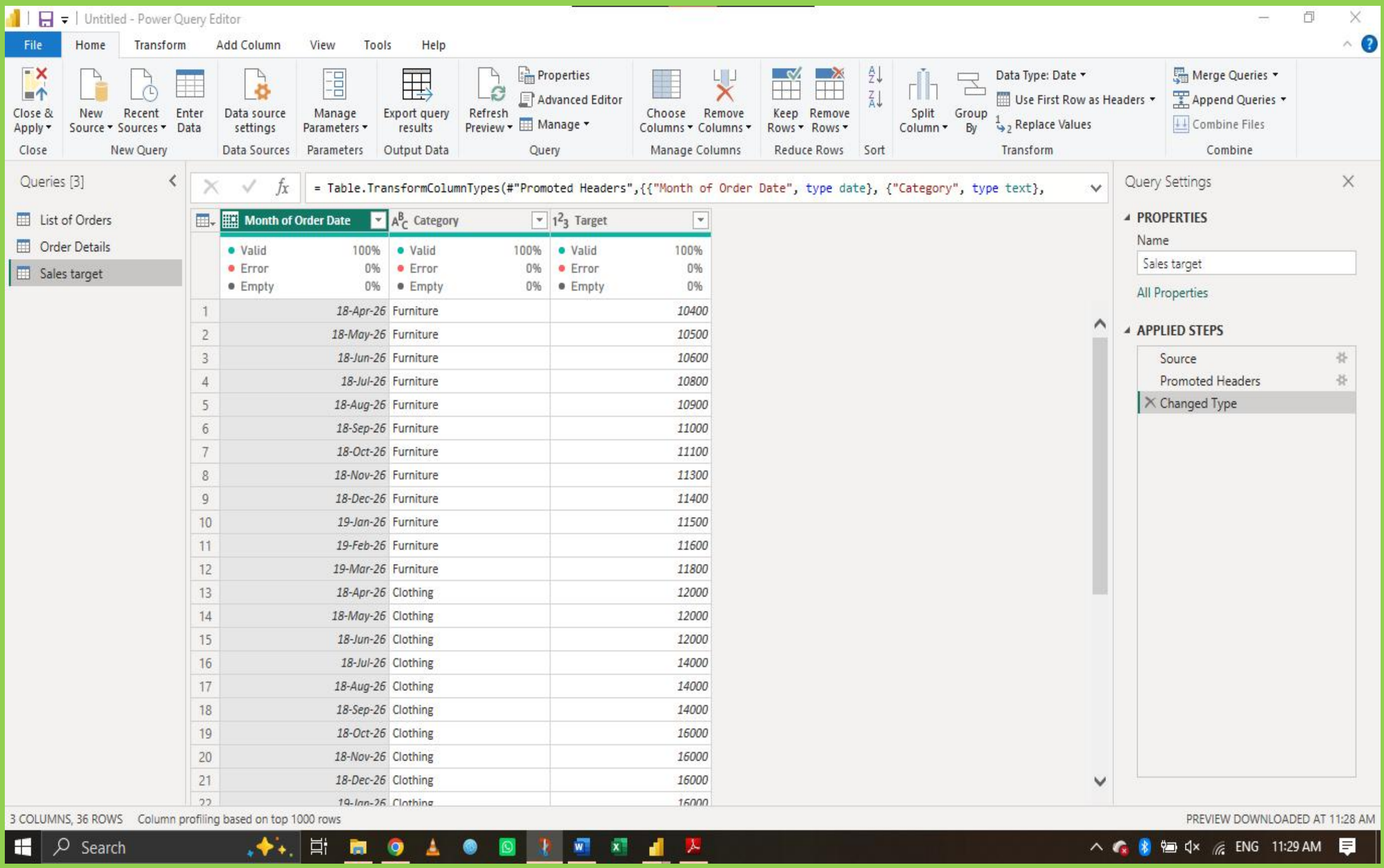
Data Import & Transformation

1. Data Import:

○ Import *List of Orders.csv* into Power BI.
○ Open *the list of orders* in **the Power Query Editor** using the **Transform Data** option.
○ Import *Order Details.csv* and *Sales Target.csv* into Power Query Editor.

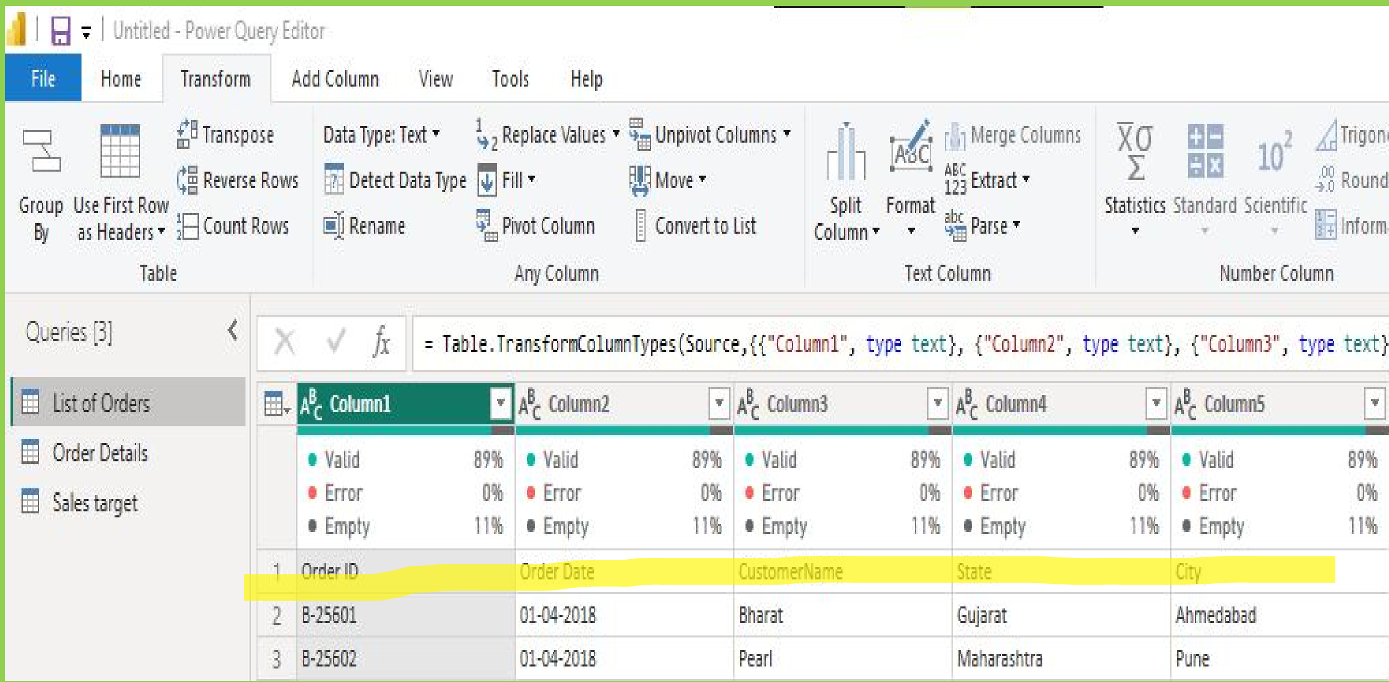


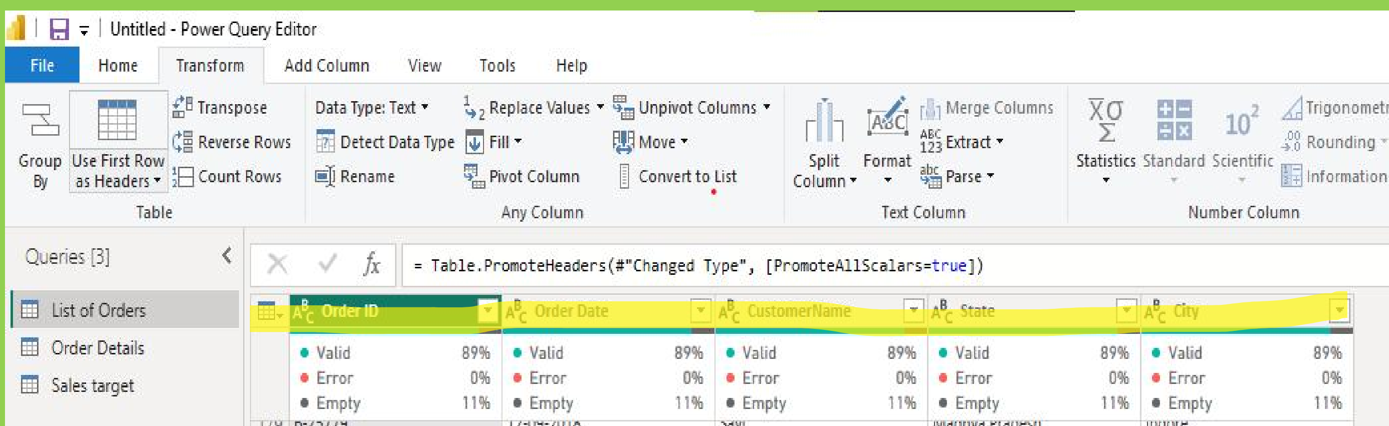


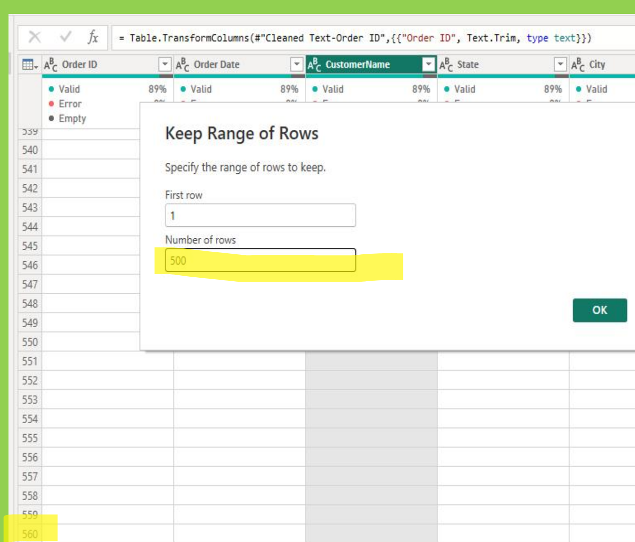


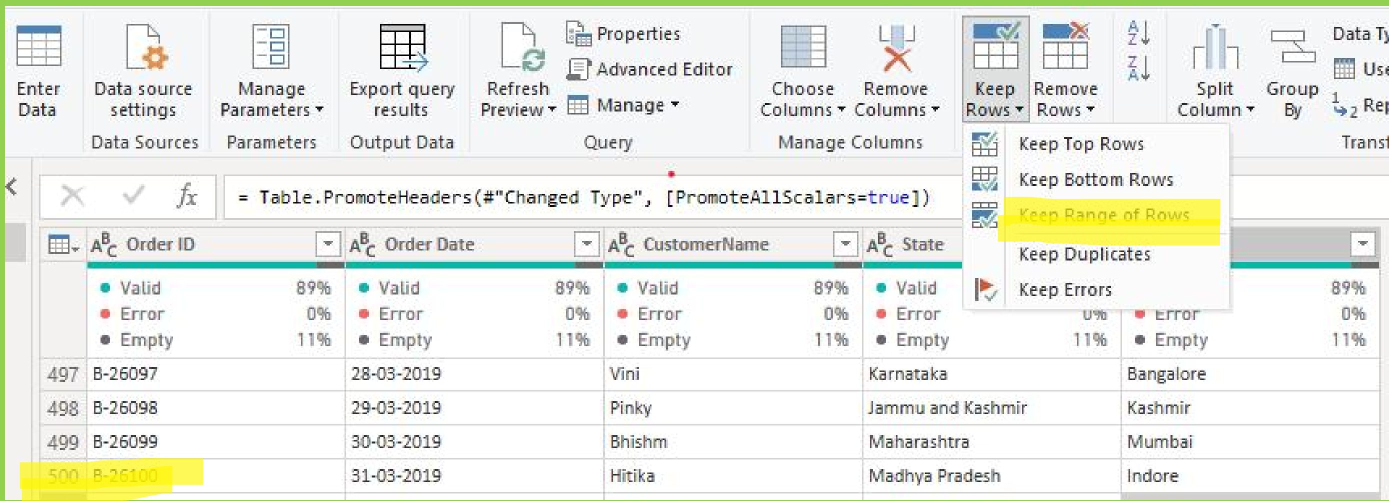
2. Row Limiting & Data Types:

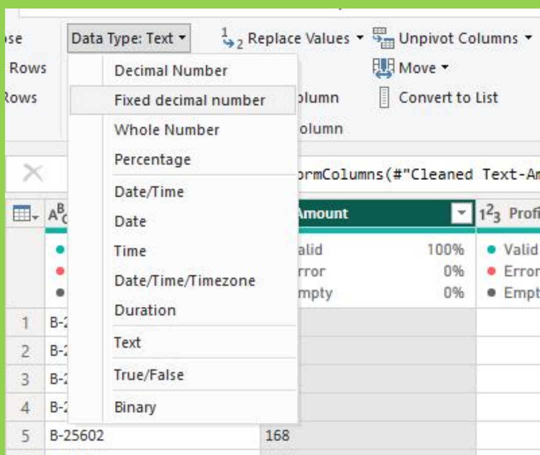
○ Restrict the *List of Orders* table to the **first 500 rows**.
○ Set the **Order Date** column to **the Date** data type.
○ Change the **Amount** and **Target** columns to **fixed decimal numbers.**

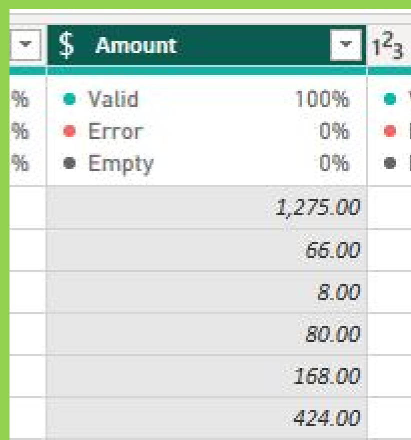

Untitled - Power Query Editor
File
Home
Transform
Add Column
View
Tools
Help
Group By
Use First Row as Headers
Transpose
Reverse Rows
Count Rows
Table
Data Type: Text
Detect Data Type
Rename
Replace Values
Fill
Pivot Column
Unpivot Columns
Move
Convert to List
Any Column
Split Column
Format
Merge Columns
Extract
Parse
Text Column
Statistics
Standard
Scientific
Number Column
Queries [3]
List of Orders
Order Details
Sales target
= Table.TransformColumnTypes(Source,{{"Column1", type text}, {"Column2", type text}, {"Column3", type text},
Column1
Column2
Column3
Column4
Column5
Valid 89%
Error 0%
Empty 11%
1 Order ID Order Date CustomerName State City
2 B-25601 01-04-2018 Bharat Gujarat Ahmedabad
3 B-25602 01-04-2018 Pearl Maharashtra Pune

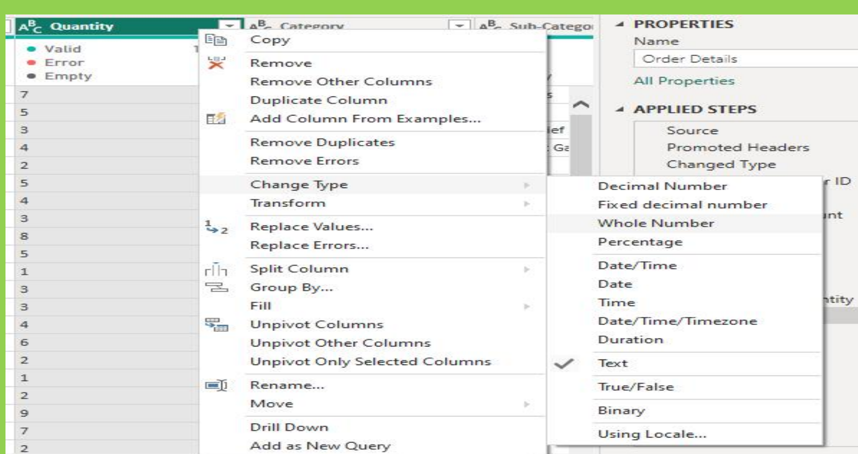

Untitled - Power Query Editor
File
Home
Transform
Add Column
View
Tools
Help
Group By
Use First Row as Headers
Transpose
Reverse Rows
Count Rows
Table
Data Type: Text
Detect Data Type
Rename
Replace Values
Fill
Pivot Column
Unpivot Columns
Move
Convert to List
Any Column
Split Column
Format
Merge Columns
Extract
Parse
Text Column
Statistics
Standard
Scientific
Trigonometry
Rounding
Information
Number Column
Queries [3]
List of Orders
Order Details
Sales target
= Table.PromoteHeaders(#"Changed Type", [PromoteAllScalars=true])
Order ID
Order Date
CustomerName
State
City
Valid 89%
Error 0%
Empty 11%

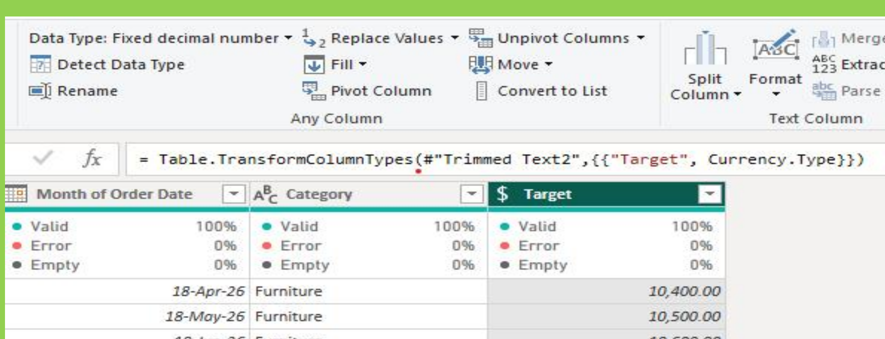

= Table.TransformColumns(#"Cleaned Text-Order ID",{{"Order ID", Text.Trim, type text}})
Order ID
Order Date
CustomerName
State
City
Keep Range of Rows
Specify the range of rows to keep.
First row
1
Number of rows
500
OK

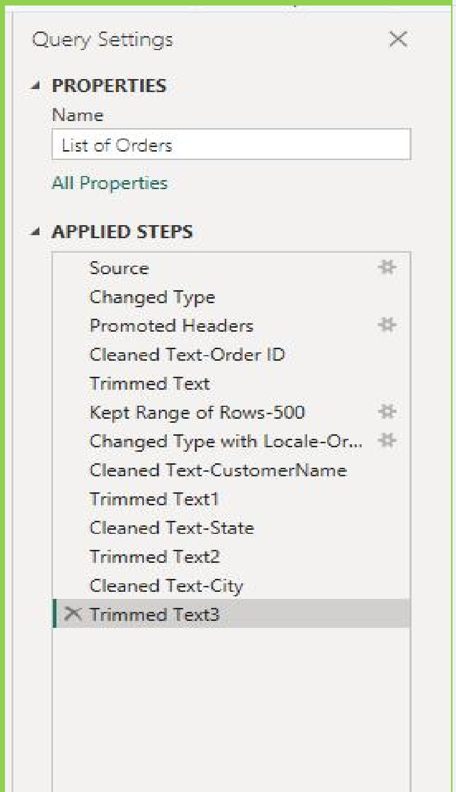

Enter Data
Data source settings
Manage Parameters
Export query results
Refresh Preview
Properties
Advanced Editor
Manage
Choose Columns
Remove Columns
Keep Rows
Remove Rows
Split Column
Group By
Data Sources
Parameters
Output Data
Query
Manage Columns
Keep Top Rows
Keep Bottom Rows
Keep Range of Rows
Keep Duplicates
Keep Errors
= Table.PromoteHeaders(#"Changed Type", [PromoteAllScalars=true])
Order ID
Order Date
CustomerName
State
497 B-26097 28-03-2019 Vini Karnataka Bangalore
498 B-26098 29-03-2019 Pinky Jammu and Kashmir Kashmir
499 B-26099 30-03-2019 Bhishm Maharashtra Mumbai
500 B-26100 31-03-2019 Hitika Madhya Pradesh Indore

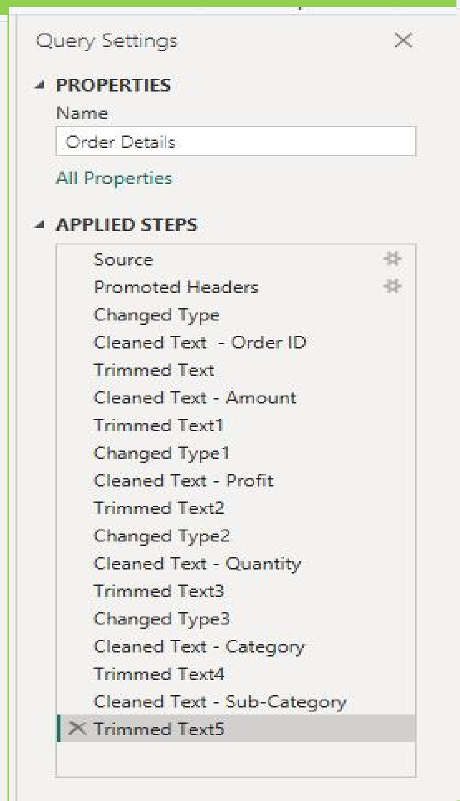

Data Type: Text
Replace Values
Unpivot Columns
Move
Convert to List
Decimal Number
Fixed decimal number
Whole Number
Percentage
Date/Time
Date
Time
Date/Time/Timezone
Duration
Text
True/False
Binary

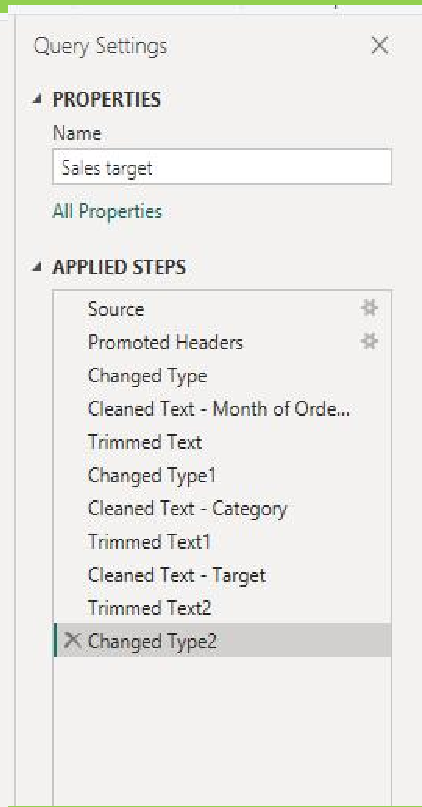

Amount
Valid 100%
Error 0%
Empty 0%
1,275.00
66.00
8.00
80.00
168.00
424.00


Quantity
Category
Copy
Remove
Remove Other Columns
Duplicate Column
Add Column From Examples...
Remove Duplicates
Remove Errors
Change Type
Transform
Replace Values...
Replace Errors...
Split Column
Group By...
Fill
Unpivot Columns
Unpivot Other Columns
Unpivot Only Selected Columns
Rename...
Move
Drill Down
Add as New Query
Decimal Number
Fixed decimal number
Whole Number
Percentage
Date/Time
Date
Time
Date/Time/Timezone
Duration
Text
True/False
Binary
Using Locale...
PROPERTIES
Name
Order Details
All Properties
APPLIED STEPS
Source
Promoted Headers
Changed Type


Data Type: Fixed decimal number
Replace Values
Unpivot Columns
Detect Data Type
Fill
Move
Rename
Pivot Column
Convert to List
Split Column
Format
Any Column
Text Column
= Table.TransformColumnTypes(#"Trimmed Text2",{{"Target", Currency.Type}})
Month of Order Date
Category
Target
18-Apr-26 Furniture 10,400.00
18-May-26 Furniture 10,500.00
18-Jun-26 Furniture 10,600.00

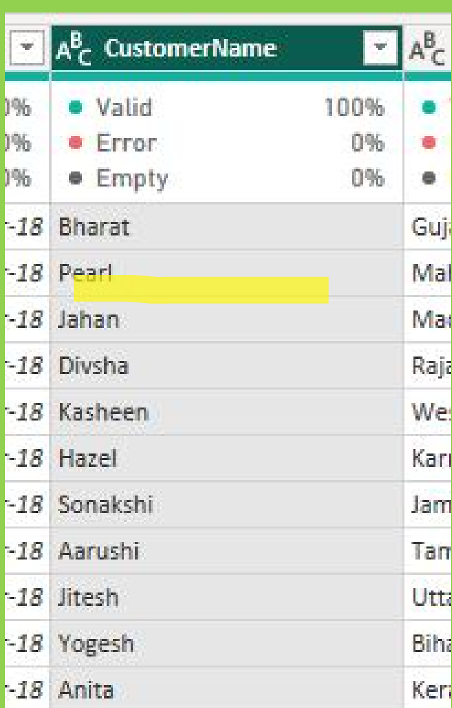


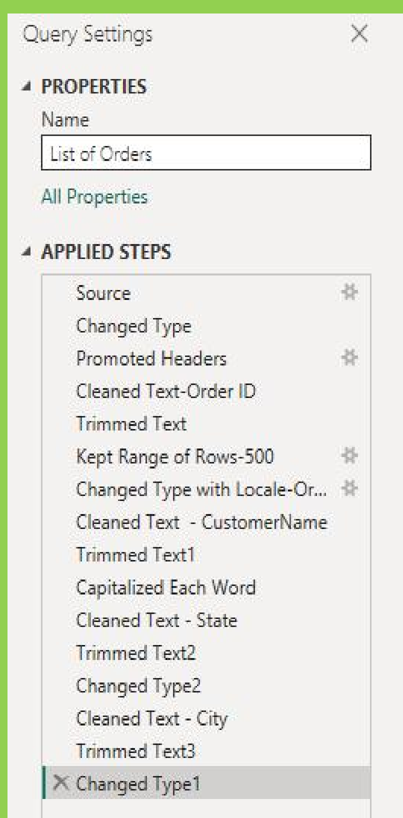




3. Text Formatting:

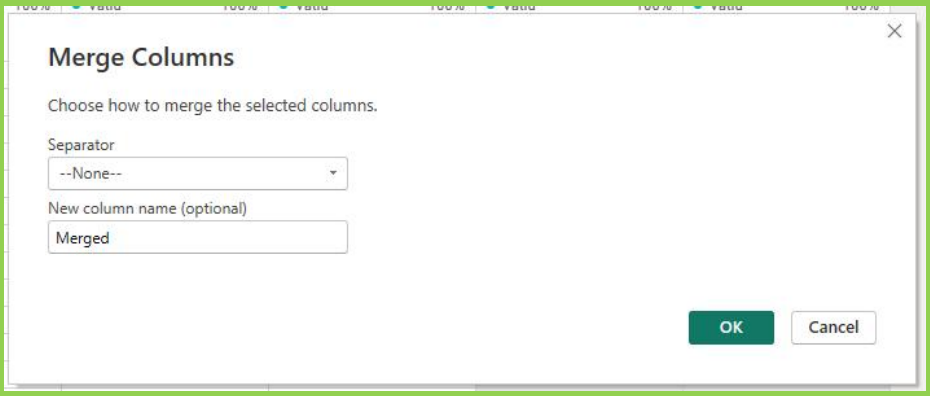
○ Format the **CustomerName** column to **proper case** to ensure consistent capitalization.

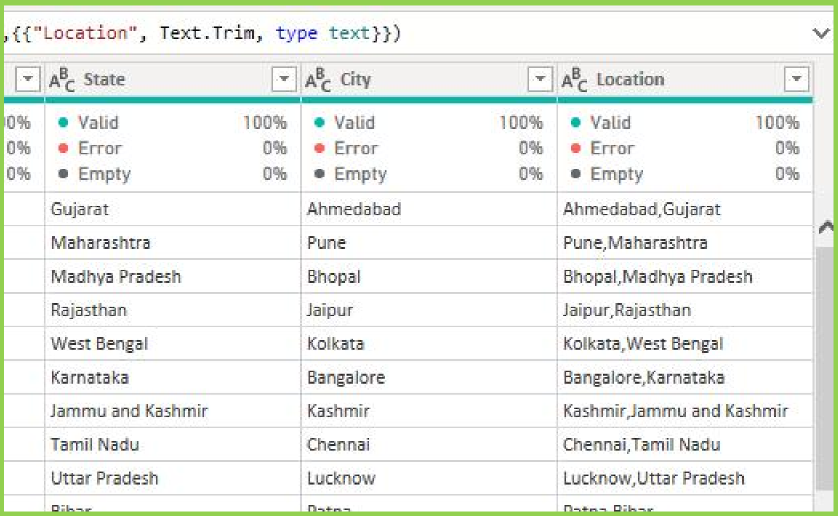




4. Column Creation:

○ Merge the **City** and **State** columns to create a new column named **Location** in the format: ***City, State***

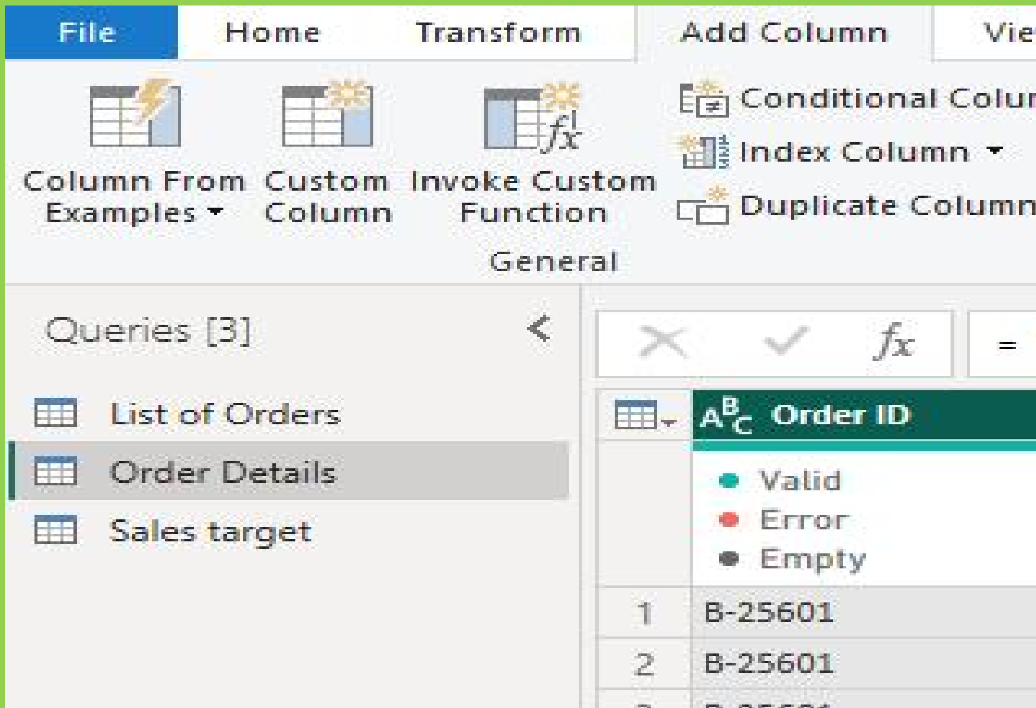


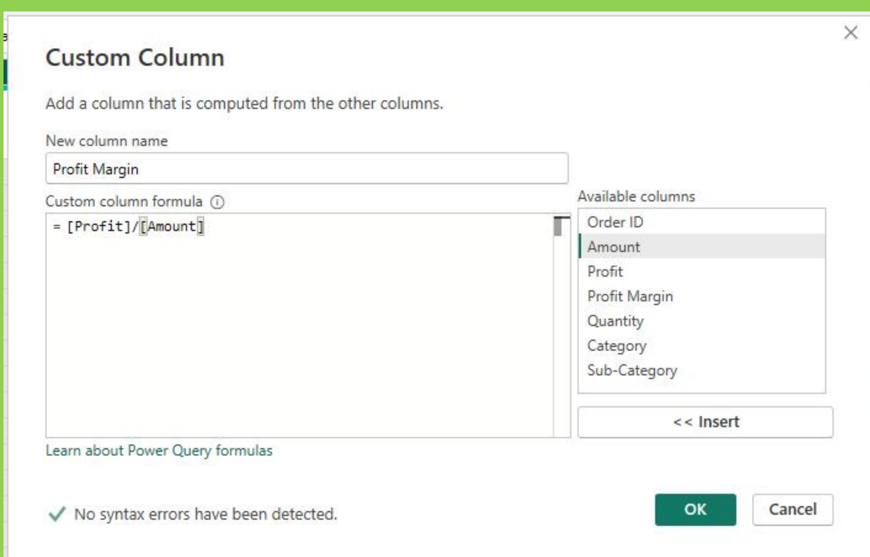


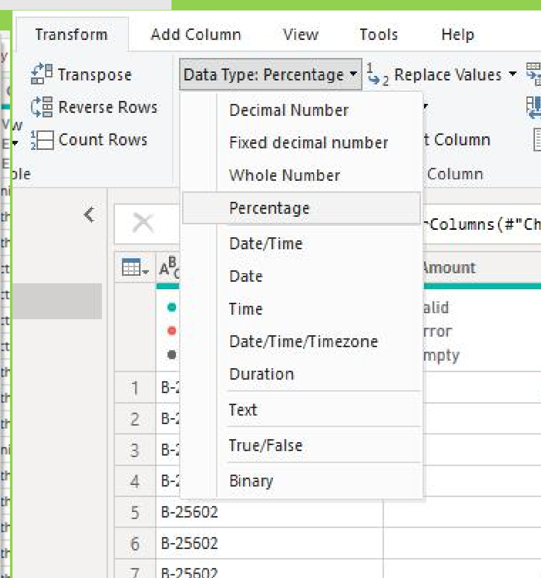
```
,{{"Location", Text.Trim, type text}})
```

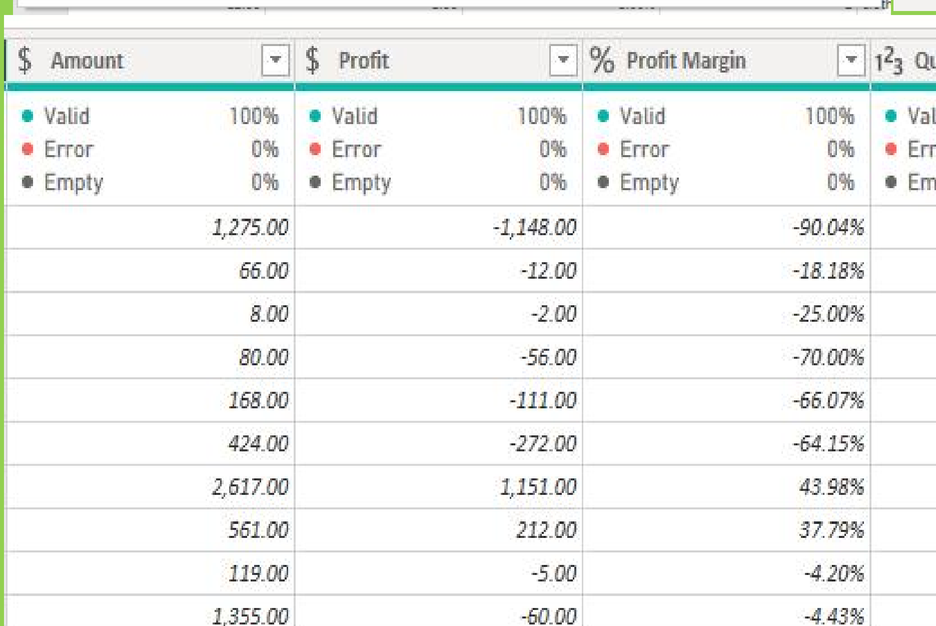
State	City	Location
Valid 100% / Error 0% / Empty 0%	Valid 100% / Error 0% / Empty 0%	Valid 100% / Error 0% / Empty 0%
Gujarat	Ahmedabad	Ahmedabad,Gujarat
Maharashtra	Pune	Pune,Maharashtra
Madhya Pradesh	Bhopal	Bhopal,Madhya Pradesh
Rajasthan	Jaipur	Jaipur,Rajasthan
West Bengal	Kolkata	Kolkata,West Bengal
Karnataka	Bangalore	Bangalore,Karnataka
Jammu and Kashmir	Kashmir	Kashmir,Jammu and Kashmir
Tamil Nadu	Chennai	Chennai,Tamil Nadu
Uttar Pradesh	Lucknow	Lucknow,Uttar Pradesh
Bihar	Patna	Patna,Bihar

○ Create a custom column named **Profit Margin** calculated as: ***Profit / Amount*** (expressed as a percentage)









$ Amount	$ Profit	% Profit Margin
Valid 100%	Valid 100%	Valid 100%
Error 0%	Error 0%	Error 0%
Empty 0%	Empty 0%	Empty 0%
1,275.00	-1,148.00	-90.04%
66.00	-12.00	-18.18%
8.00	-2.00	-25.00%
80.00	-56.00	-70.00%
168.00	-111.00	-66.07%
424.00	-272.00	-64.15%
2,617.00	1,151.00	43.98%
561.00	212.00	37.79%
119.00	-5.00	-4.20%
1,355.00	-60.00	-4.43%

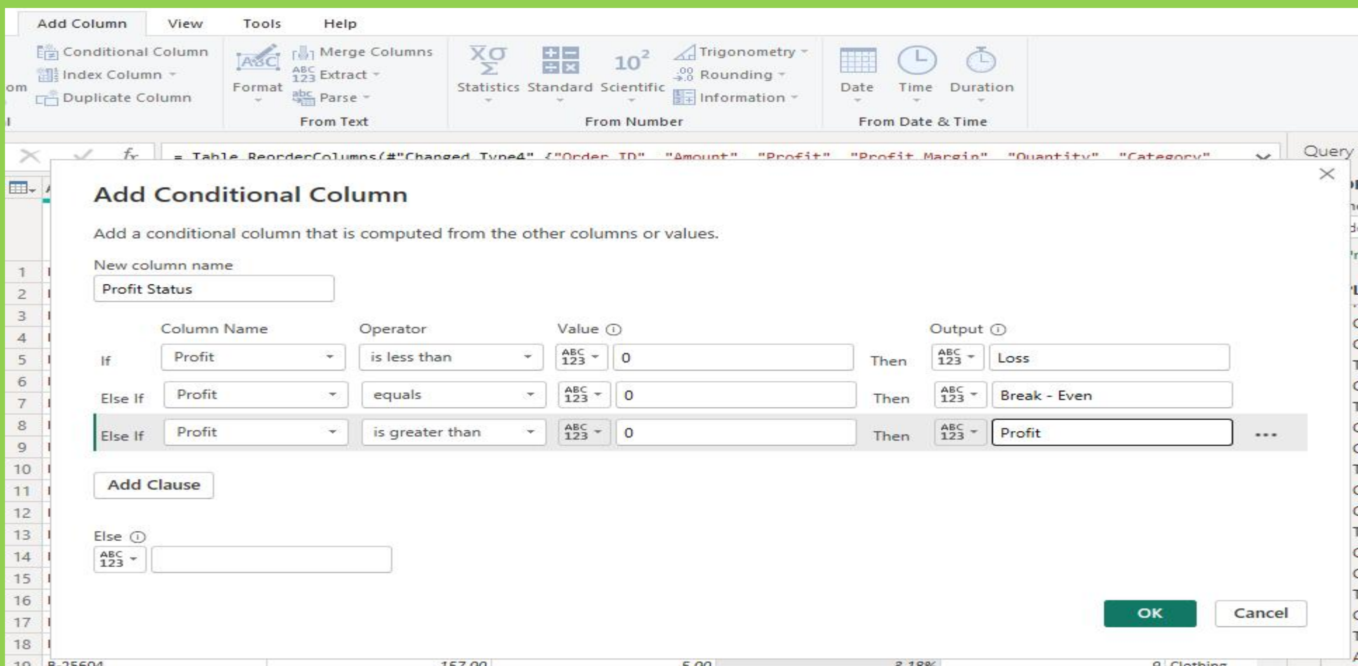
5. Conditional Column:

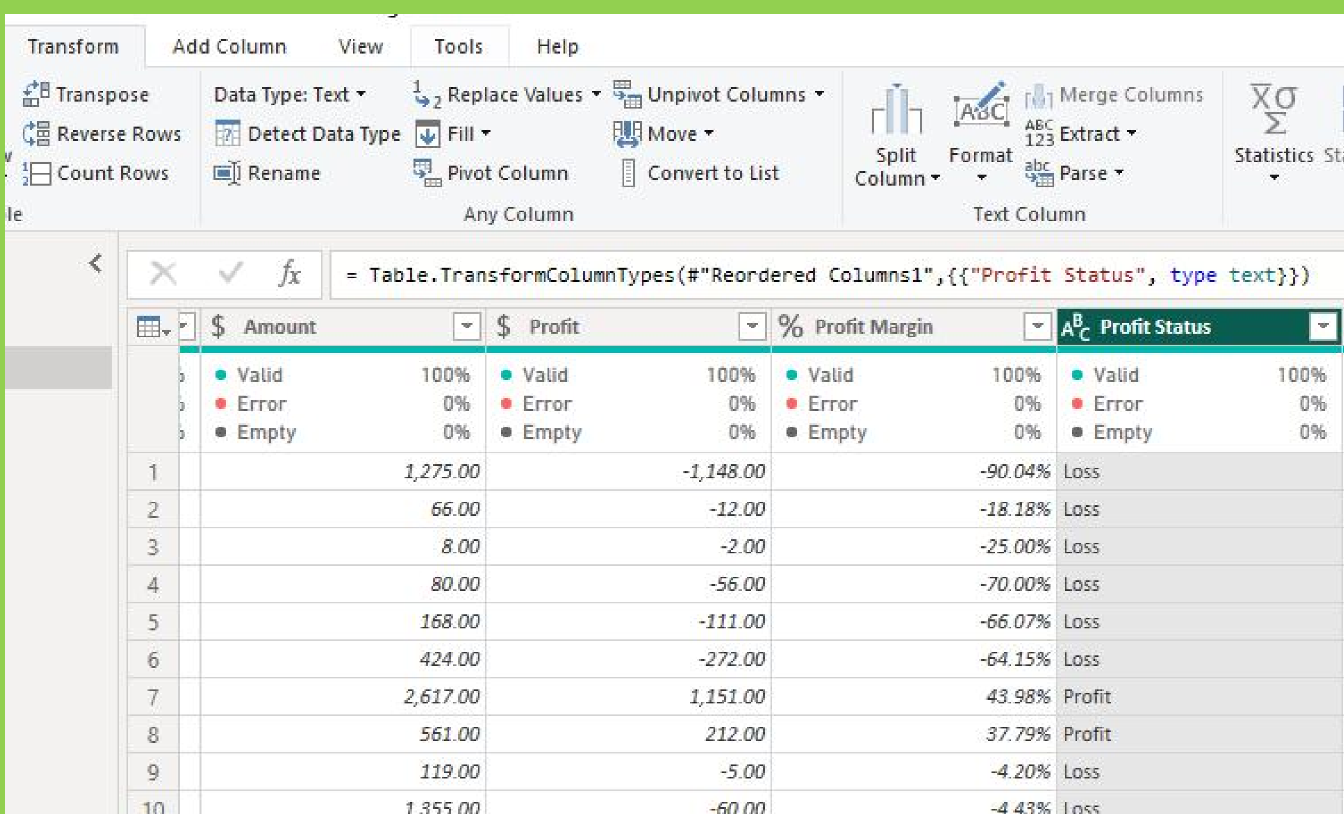
○ Add a conditional column named **Profit Status** based on:

■ Profit < 0 → *Loss*

■ Profit = 0 → *Break-Even*

■ Profit > 0 → *Profit*

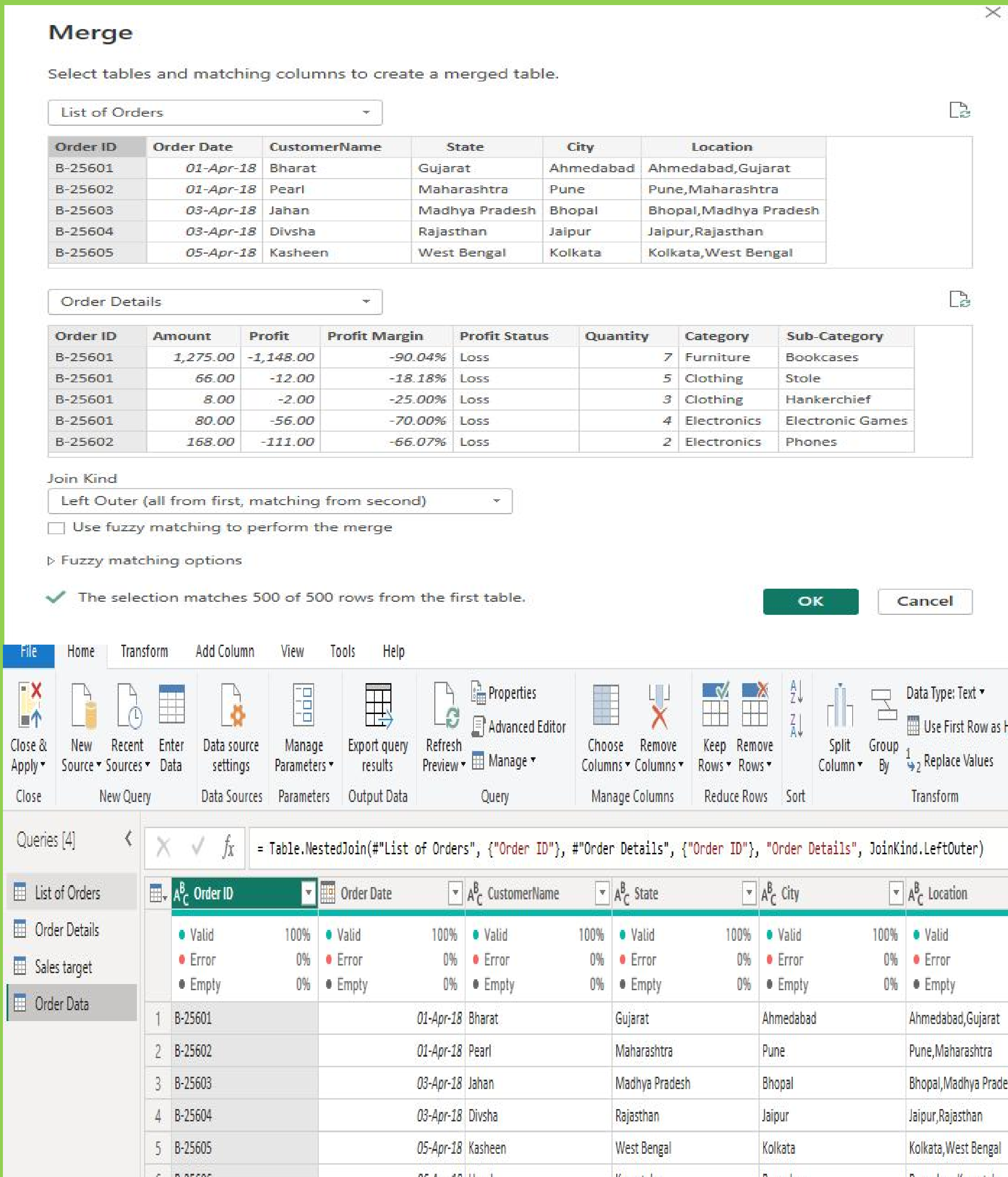




6. Merging Data (Joins)

○ Merge the *List of Orders* and *Order Details* tables using **Order ID**.

○ Name the merged output table as **Orders Data**.



7. Handling Missing Data & Duplicate Data

● Identify missing values across all tables.
● Define and apply appropriate strategies such as:
○ Replacing missing values
○ Removing records
○ Retaining nulls with justification
● Identify duplicate rows and define a suitable handling strategy based on business logic.

For Handling Missing Data & Duplicate Data in Power BI (Power Query Editor), you can document and perform the cleaning process like this:

Handling Missing Data & Duplicate Data
1. Identify Missing Values Across All Tables
Steps:
Open Power BI Desktop
Go to Home → Transform Data
Select each table one by one
Review columns for:
null
blank values
empty cells

You can also:

Go to View → Column Quality
Enable Column Distribution and Column Profile

These help identify missing values quickly.

2. Apply Missing Value Handling Strategy
A. Replace Missing Values

Use when values can reasonably be defaulted.

Example:

Discount Column

Replace null → 0

Steps:

Select column
Transform → Replace Values

Replace:

null → 0
Example:

Quantity Column

Replace null → 1

Reason:
Assume minimum order quantity.

Steps:

Select column
Transform → Replace Values
null → 1
B. Remove Records

Use when critical information is missing.

Example:
If Order ID is missing:

Steps:

Select Order ID
Home → Remove Rows → Remove Blank Rows

Reason:
Records without unique identifiers cannot be matched or analyzed.

C. Retain Null Values (with justification)

Use when missing data carries meaning.

Example:
Customer Feedback

Keep null values

Reason:
Blank may indicate customer did not provide feedback.

No action required.

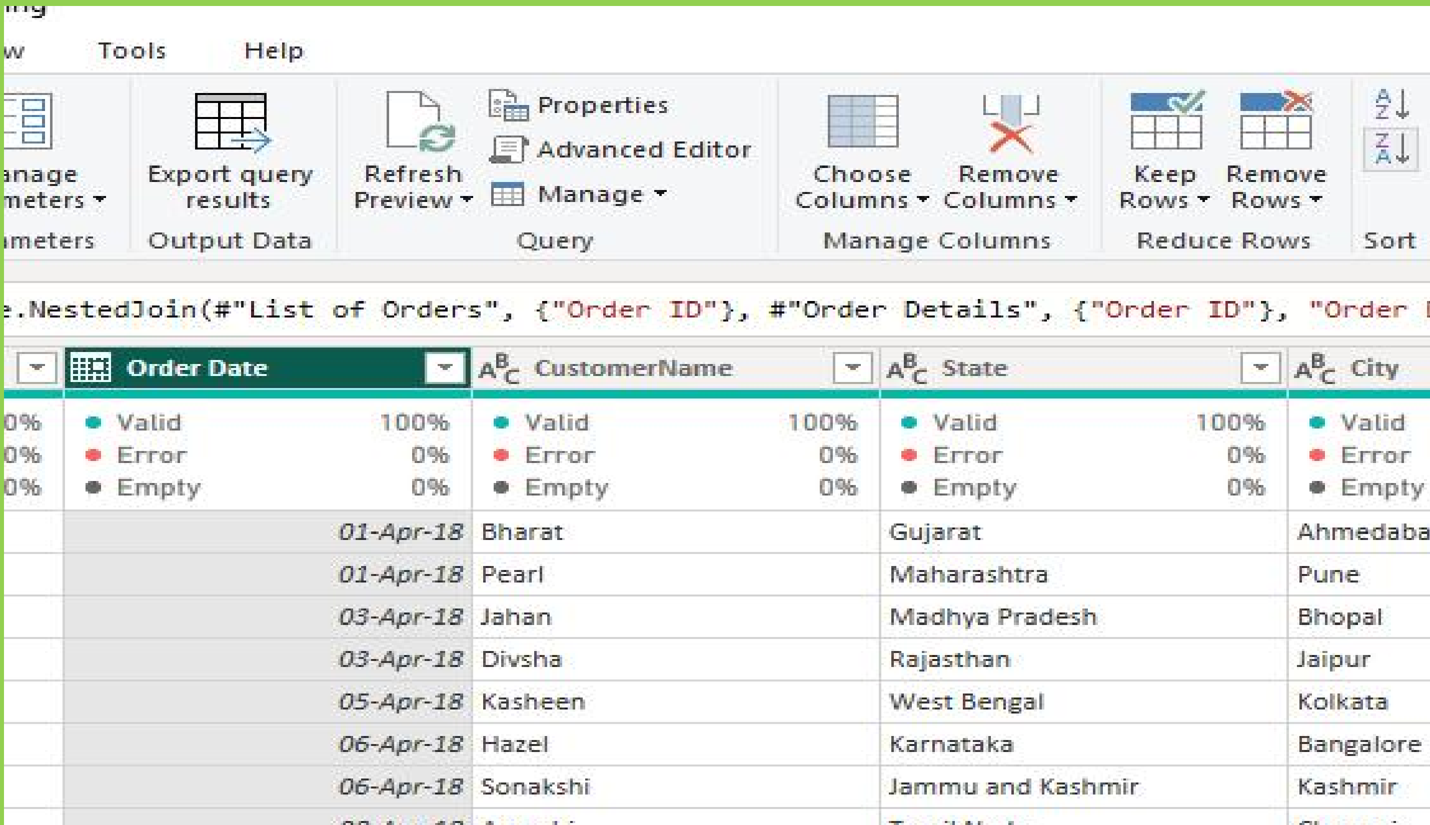
3. Identify Duplicate Rows
Steps:
Select table
Select all columns or key columns
Go to:
Home → Remove Rows → Remove Duplicates

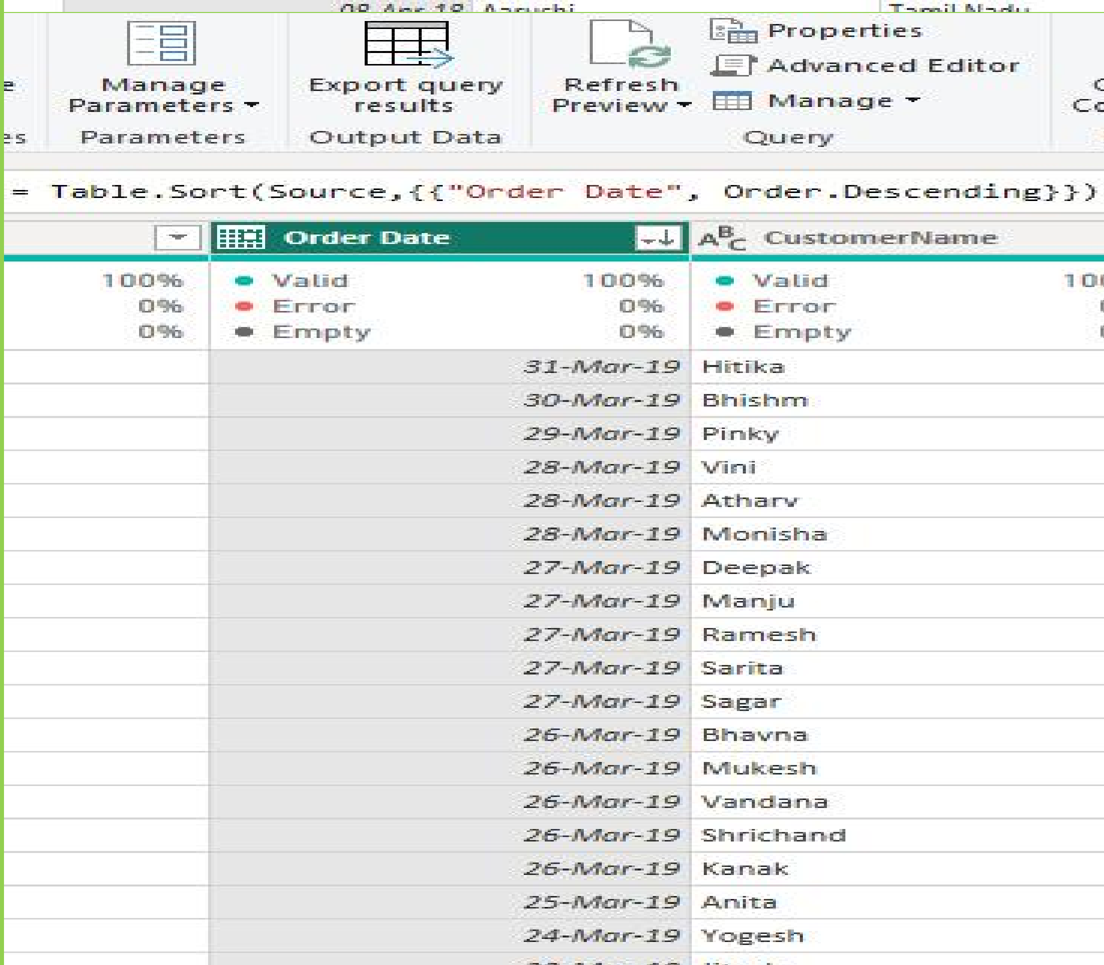
Suggested Documentation Statement

Missing values were identified using Column Quality and handled using replacement, removal, or retention strategies depending on business relevance. Duplicate rows were evaluated using business rules and removed only when they represented unintended repeated records.

8. Sorting and Filtering Data

● Sort records in the **Orders Data** table by **Order Date** in **descending order**.



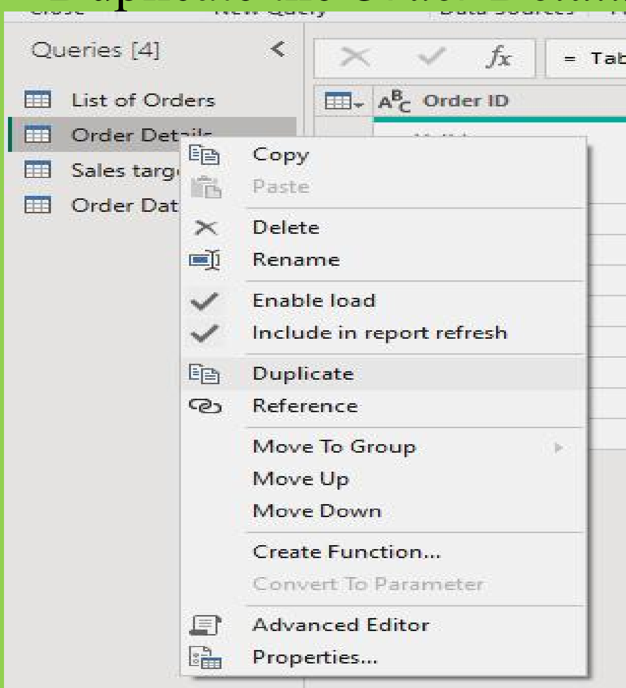


● Apply filters to analyze orders from a specific state (e.g., **Tamil Nadu**).

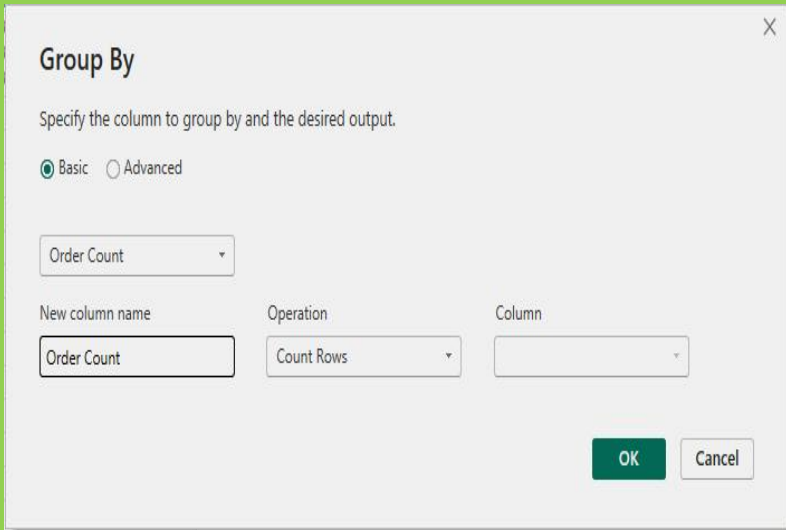
	Order ID	Order Date	CustomerName	State	City	Location
	Valid 100%	Valid 100%	Valid 100%	Valid 100%	Valid 100%	Valid 10
	Error 0%	Error 0%	Error 0%	Error 0%	Error 0%	Error
	Empty 0%	Empty 0%	Empty 0%	Empty 0%	Empty 0%	Empty
1	B-26081	22-Mar-19	Aarushi	Tamil Nadu	Chennai	Chennai,Tamil Nadu
2	B-26018	14-Feb-19	Aarushi	Tamil Nadu	Chennai	Chennai,Tamil Nadu
3	B-26008	09-Feb-19	Kalyani	Tamil Nadu	Chennai	Chennai,Tamil Nadu
4	B-25860	15-Nov-18	Akshay	Tamil Nadu	Chennai	Chennai,Tamil Nadu
5	B-25788	21-Sep-18	Dinesh	Tamil Nadu	Chennai	Chennai,Tamil Nadu
6	B-25716	11-Jul-18	Surabhi	Tamil Nadu	Chennai	Chennai,Tamil Nadu
7	B-25698	23-Jun-18	Amisha	Tamil Nadu	Chennai	Chennai,Tamil Nadu
8	B-25608	08-Apr-18	Aarushi	Tamil Nadu	Chennai	Chennai,Tamil Nadu

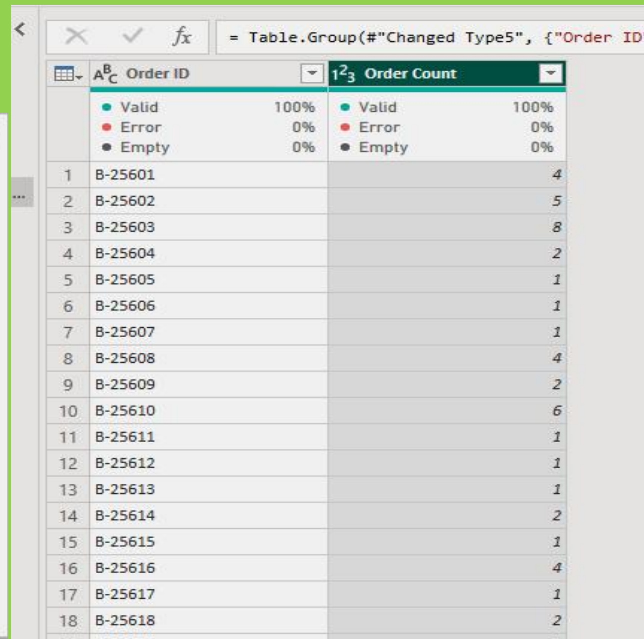
9. Grouping and Aggregating Data

● Duplicate the *Order Details* table and perform the following:



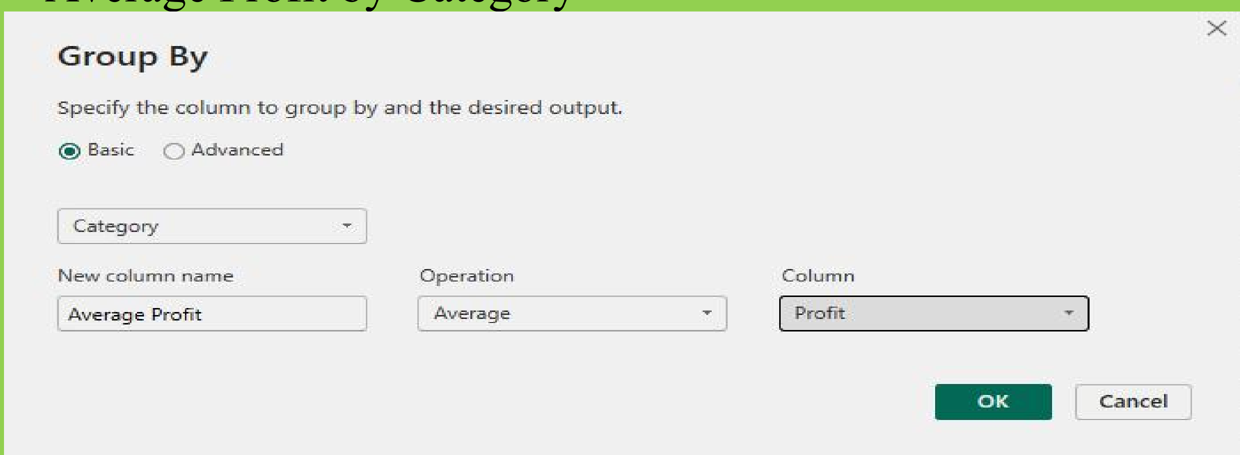
○ Count of Order ID

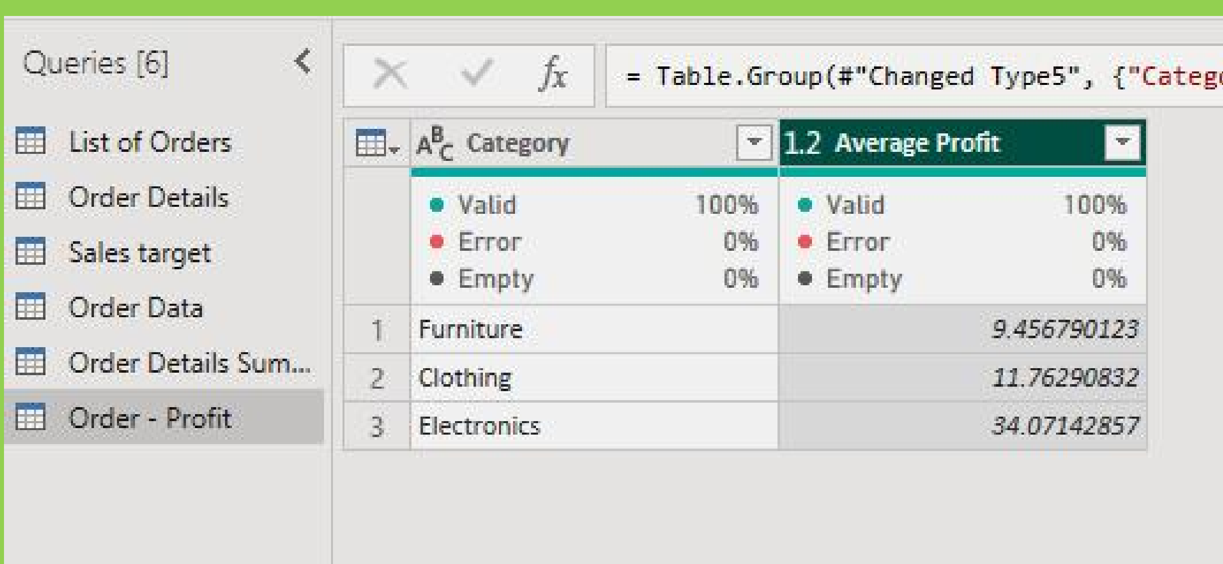




	Order ID	Order Count
	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%
1	B-25601	4
2	B-25602	5
3	B-25603	8
4	B-25604	2
5	B-25605	1
6	B-25606	1
7	B-25607	1
8	B-25608	4
9	B-25609	2
10	B-25610	6
11	B-25611	1
12	B-25612	1
13	B-25613	1
14	B-25614	2
15	B-25615	1
16	B-25616	4
17	B-25617	1
18	B-25618	2

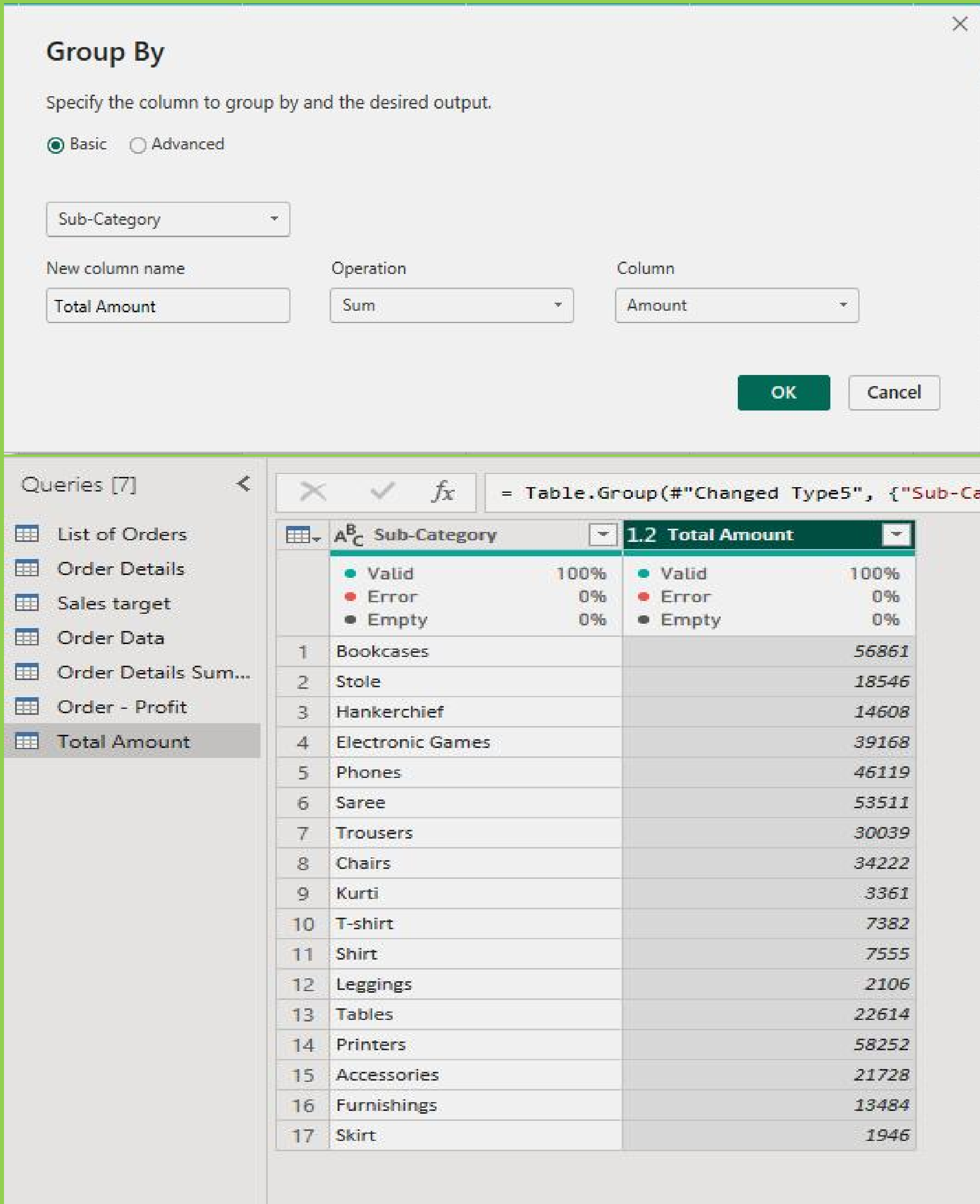
○ Average Profit by Category



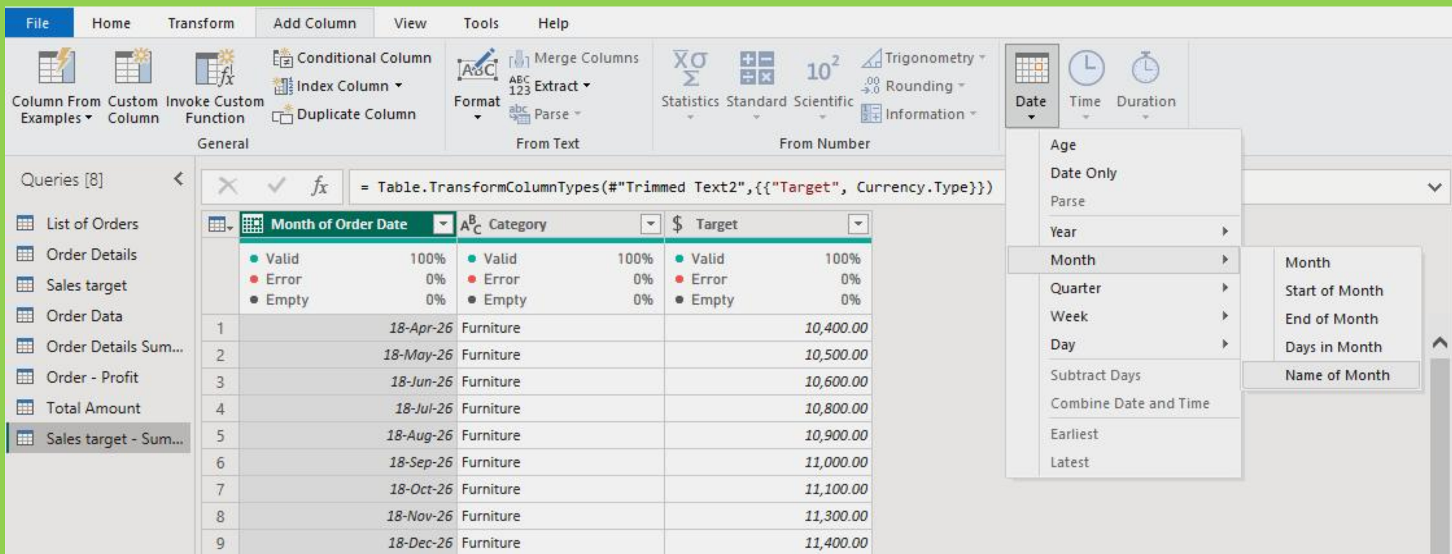


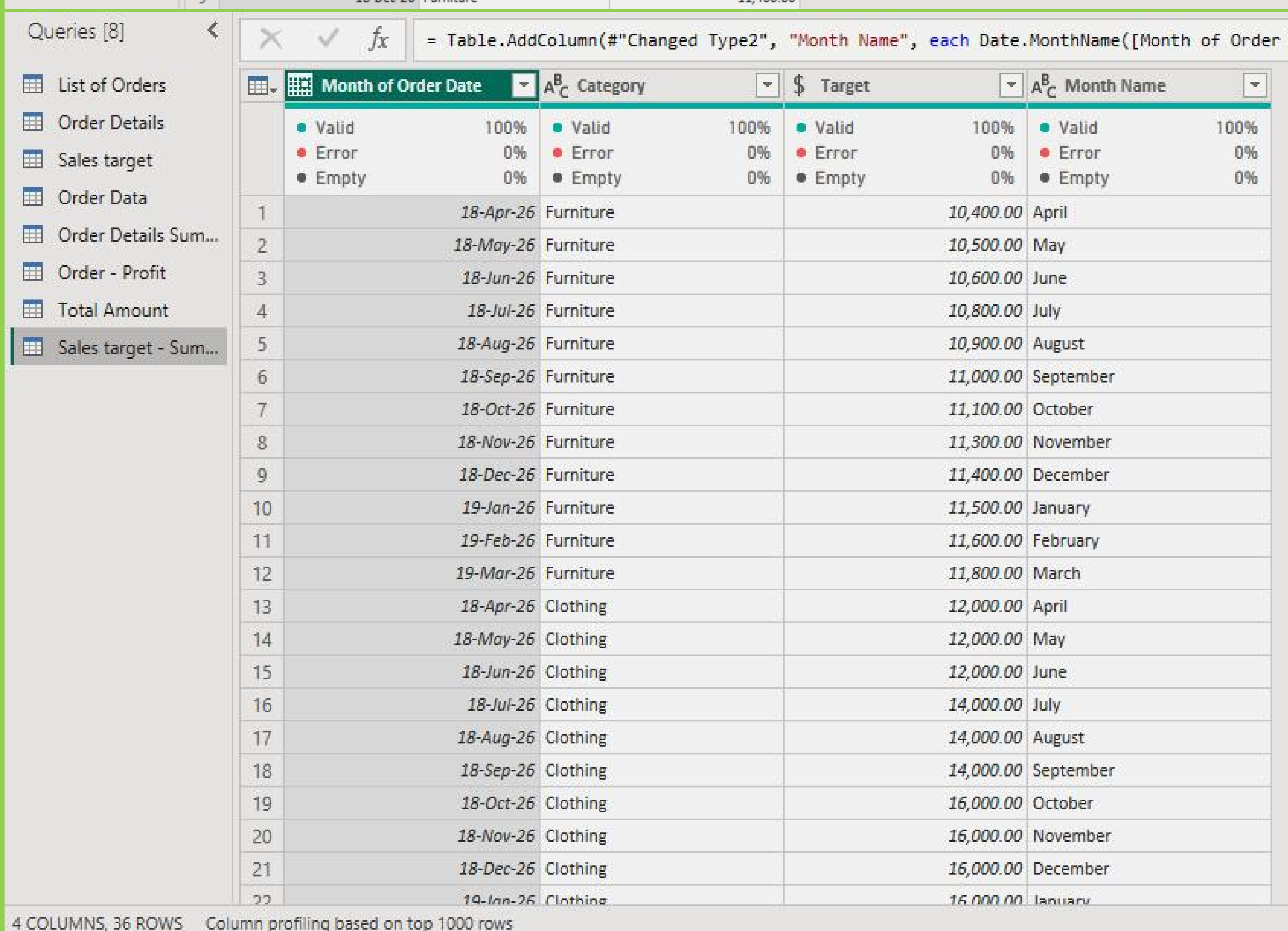
	Category	Average Profit
	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%
1	Furniture	9.456790123
2	Clothing	11.76290832
3	Electronics	34.07142857

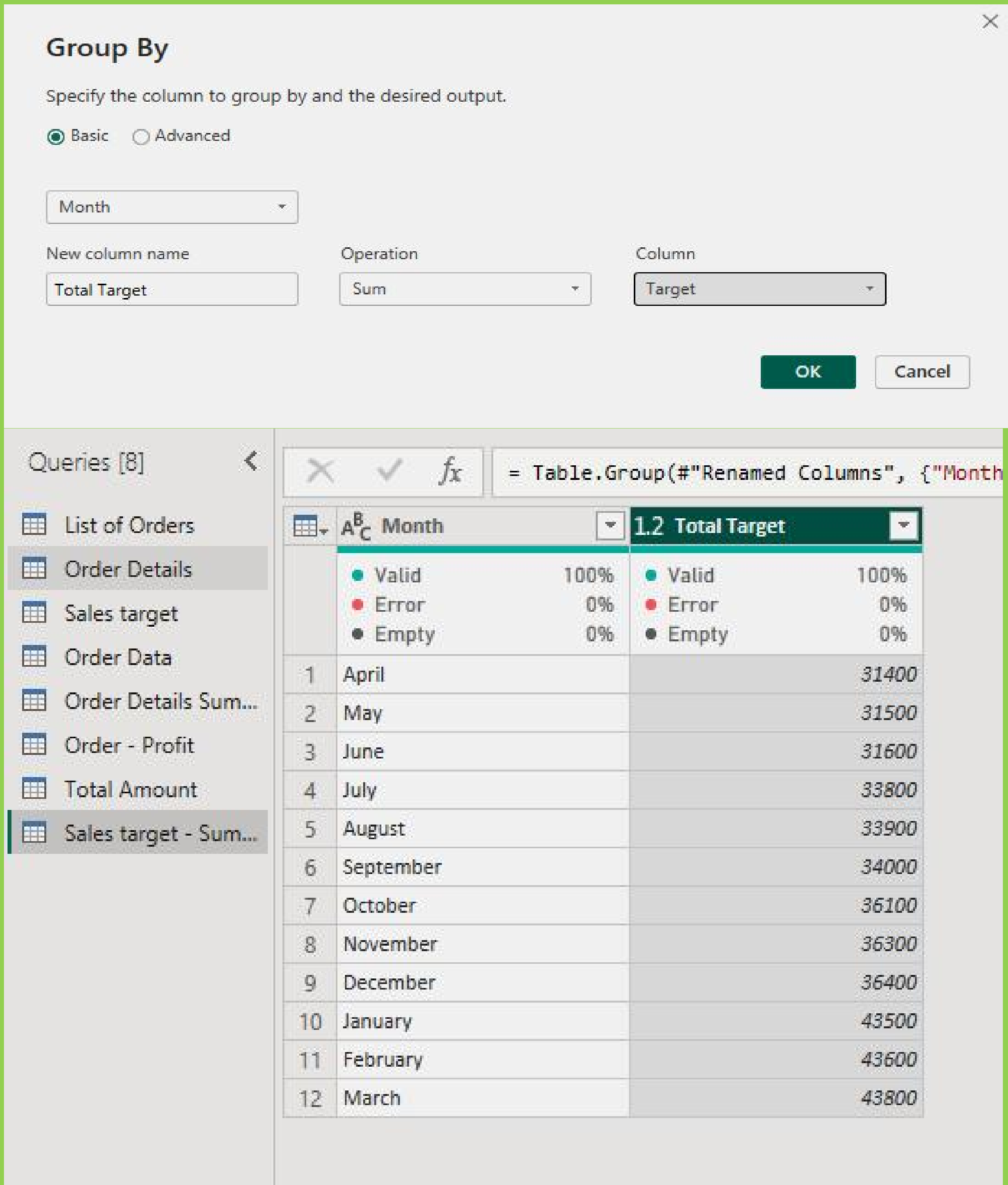
○ Total Amount by Sub-Category



● Duplicate the *Sales Target* table and aggregate the total Target by **Month of order date.**



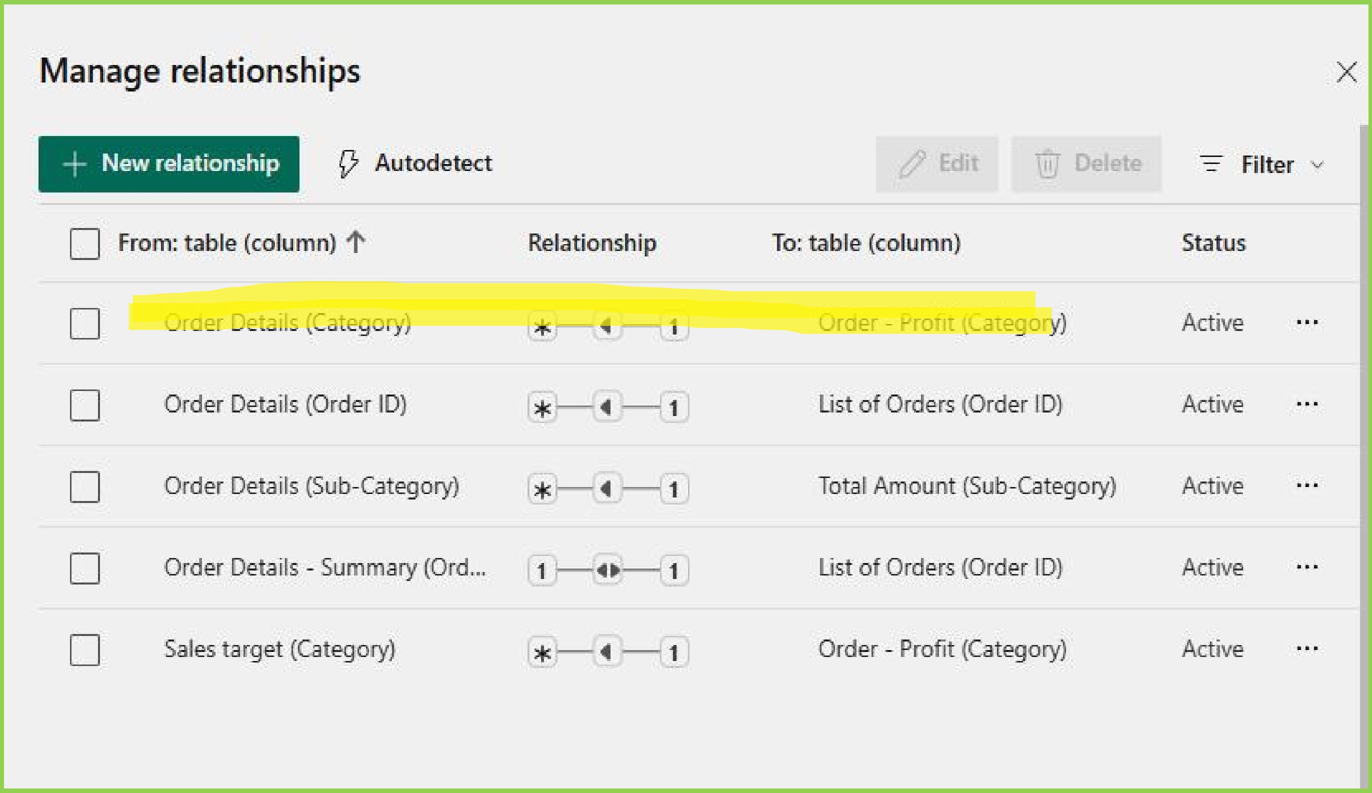



Group By
Specify the column to group by and the desired output.
Basic
Advanced
Month
New column name
Total Target
Operation
Sum
Column
Target
OK
Cancel
Queries [8]
List of Orders
Order Details
Sales target
Order Data
Order Details Sum...
Order - Profit
Total Amount
Sales target - Sum...
= Table.Group(#"Renamed Columns", {"Month
Month
1.2 Total Target
Valid 100%
Error 0%
Empty 0%
Valid 100%
Error 0%
Empty 0%
1 April 31400
2 May 31500
3 June 31600
4 July 33800
5 August 33900
6 September 34000
7 October 36100
8 November 36300
9 December 36400
10 January 43500
11 February 43600
12 March 43800

10. Data Modeling:

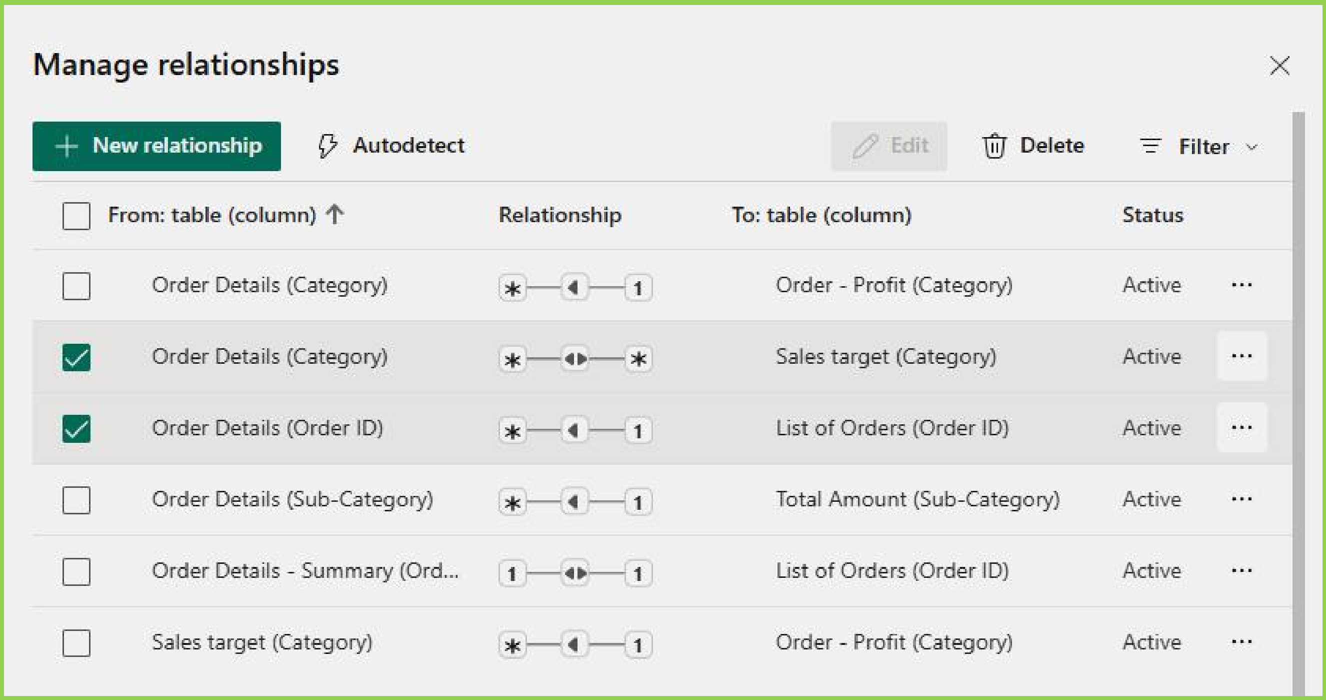
● Establish a relationship between:

○ *List of Orders* and *Order Details* using **Order ID**



● Establish a relationship between:

○ *Order Details* and *Sales Target* using **Category**



2. Use **Manage Relationships** to ensure relationships are active and correctly defined.